

Saddle-Point Approximation and Functional Determinants

Source: Perimeter Scholars International, Quantum Field Theory II, Tutorial 3; 15 worked parts.

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Question prompts paraphrased from the official Perimeter PSI Online tutorial sheet; the equations and notation follow the source.

1 Saddle-Point Approximation

Approximation methods for ordinary integrals are among the reasons that path integrals are useful. For example, consider

$$I(z) = \int dx e^{if(x,z)}. \quad (1.1)$$

The corresponding ideas extend to functional integrals such as

$$Z[J] = \int \mathcal{D}\phi e^{iS[\phi,J]}. \quad (1.2)$$

The saddle-point method is one especially useful example.

Restore $\hbar$ in the Euclidean partition function:

$$Z[J] = \int \mathcal{D}\phi e^{-\frac{1}{\hbar}S[\phi,J]}. \quad (1.3)$$

This makes it possible to organise a semiclassical expansion by treating $\hbar$ as small. In Euclidean signature, configurations ϕ_c minimising the action dominate because configurations with larger action are exponentially suppressed. For a Minkowski integral, all stationary configurations of the action must be considered rather than only its minima.

The following questions first develop this approximation for ordinary n -dimensional integrals.

1. Let $S(x^\mu; z)$ be a real function of n spacetime variables $\mu = 1, \dots, n$ and a coupling z , and define

$$I(z) = \int_{-\infty}^{\infty} d^n x e^{-\frac{1}{\hbar} S(x;z)}. \quad (1.4)$$

Expand $S(x; z)$ around its extrema x_c and show that the sum over its minima gives

$$I(z) \approx \sum_{\text{Min}[x_c]} e^{-\frac{1}{\hbar} S(x_c; z)} \left[\frac{(2\pi\hbar)^n}{\det S''(x_c)} \right]^{1/2} + \mathcal{O}(\hbar^a), \quad (1.5)$$

where

$$[S''(x_c)]_{\mu\nu} = \left. \frac{\partial^2 S}{\partial x^\mu \partial x^\nu} \right|_{x=x_c}. \quad (1.6)$$

Solution. Expanding $S(x; z)$ around its minima, we find

$$S(x; z) = S(x_c; z) + \frac{1}{2} (x - x_c)^\mu (x - x_c)^\nu [S''(x_c; z)]_{\mu\nu} + \dots \quad (1.7)$$

In the semi-classical regime, the field configurations around the minima dominate, giving

$$\begin{aligned} I(z) &\approx \sum_{\text{Min}[x_c]} \int d^n x e^{-\frac{1}{\hbar} (S(x_c; z) + \frac{1}{2} (x - x_c)^\mu (x - x_c)^\nu [S''(x_c; z)]_{\mu\nu})} \\ &= \sum_{\text{Min}[x_c]} e^{-\frac{1}{\hbar} S(x_c; z)} \left[\frac{(2\pi\hbar)^n}{\det S''(x_c)} \right]^{1/2}, \end{aligned} \quad (1.8)$$

where we have used the results from conventional Gaussian integration in the last equality.

2. For a generic function $S(x; z)$, determine the exponent a .

Solution. The important fluctuations should be $x - x_c \sim \sqrt{\hbar}$. The next contribution comes from S'''' (S''' is odd, and hence vanishes). Therefore, $a = n/2 + 2 - 1 = n/2 + 1$.

3. State the condition under which the approximation becomes exact.

Solution. The approximation is exact if all higher order derivative vanishes.

4. Explain how the saddle-point expression should change if the starting point is the Minkowski partition function

$$Z = \int e^{iS/\hbar}. \quad (1.9)$$

Solution. In real time, $-S \rightarrow iS$. Besides that, we also have to add the $i\epsilon$ prescription to ensure the path integral converges. Therefore, we have

$$I(z) \approx \sum_{\text{Min}[x_c]} e^{\frac{i}{\hbar} S(x_c; z)} \left[\frac{(2\pi i \hbar)^n}{\det(S''(x_c) + i\epsilon)} \right]^{1/2} + \mathcal{O}(\hbar^a). \quad (1.10)$$

2 Functional Determinants

Functional determinants occur throughout the path-integral treatment of QFT, including the semiclassical result above. Their formal definition as a product over operator eigenvalues is generally divergent and requires regularisation. For this exercise, set those analytic subtleties aside and manipulate the operators by analogy with matrices.

Define

$$F[f] = \log \left(\det \left[-\frac{d^2}{dy^2} + f(y) \right] \right). \quad (2.1)$$

Thus F is the logarithm of the determinant of the operator

$$\mathcal{A}[f(y)] = -\frac{d^2}{dy^2} + f(y). \quad (2.2)$$

If $M(y, z)$ is the integral kernel of an operator $\mathcal{M}$, then

$$\mathcal{M}g(y) = \int M(y, z)g(z) dz \quad (2.3)$$

for every suitable function g .

The matrix-product analogy used here is

$$\sum_y M_{xy} M_{yz} \iff \int dy M(x, y) M(y, z). \quad (2.4)$$

1. Show that the integral kernel of $\mathcal{A}$ is

$$A(y, z) = \left[-\frac{d^2}{dy^2} + f(y) \right] \delta(y - z). \quad (2.5)$$

Solution.

$$\mathcal{A}[f(y)]g(y) = -\frac{d^2 g}{dy^2} + fg = \int dz A(y, z)g(z). \quad (2.6)$$

Suppose that $\mathcal{O}(x, y)$ is the integral kernel of

$$\mathcal{O} = \left(-\frac{d^2}{dy^2} + f(y) \right)^{-1}. \quad (2.7)$$

2. The operators satisfy $\mathcal{A}\mathcal{O} = 1$. Rewrite this identity entirely in terms of their integral kernels.

Solution.

$$\int dz A(x, z) O(z, y) = \delta(x - y). \quad (2.8)$$

3. Using functional differentiation, prove

$$\frac{\delta F[f]}{\delta f(x)} = O(x, x). \quad (2.9)$$

Hint: Recall the matrix identity

$$\det M = \exp(\text{tr} \log M). \quad (2.10)$$

Solution.

$$\frac{\delta F[f]}{\delta f(x)} = \frac{\delta}{\delta f(x)} \text{tr} \log \mathcal{A} = \frac{\delta}{\delta f(x)} \text{tr} \mathcal{O} \delta \mathcal{A} = \frac{\delta}{\delta f(z)} \int dy dz O(x, y) \delta f(y) \delta(y - z) = O(x, x). \quad (2.11)$$

4. Next show

$$\frac{\delta^2 F[f]}{\delta f(x) \delta f(y)} = -O(x, y) O(y, x). \quad (2.12)$$

Hint: To differentiate an inverse, begin with

$$M^{-1} M = 1. \quad (2.13)$$

Solution.

$$\frac{\delta^2 F[f]}{\delta f(x) \delta f(y)} = \frac{\delta}{\delta f(y)} \text{tr} \mathcal{A}^{-1} = -\text{tr} \mathcal{A}^{-1} \frac{\delta \mathcal{A}}{\delta f(y)} \mathcal{A}^{-1} = -O(x, y) O(y, x). \quad (2.14)$$

5. Finally derive

$$\begin{aligned} \frac{\delta^3 F[f]}{\delta f(x) \delta f(y) \delta f(z)} &= O(x, y) O(y, z) O(z, x) \\ &\quad + O(x, z) O(z, y) O(y, x). \end{aligned} \quad (2.15)$$

Solution. Follows trivially from above.

3 Optional: The Gel'fand--Yaglom Formalism

The Gel'fand--Yaglom method can determine functional determinants without explicitly computing every eigenvalue. In favourable cases it is exact, and it can also be implemented numerically. Begin with the Riemann zeta function

$$\zeta(s) \equiv \sum_{k=1}^{\infty} k^{-s}. \quad (3.1)$$

The contour required below is fixed by the argument principle. It is positively oriented and surrounds the positive zeros λ_n of $F(\lambda)$, while avoiding the branch cut of λ^{-s} on the negative real axis.

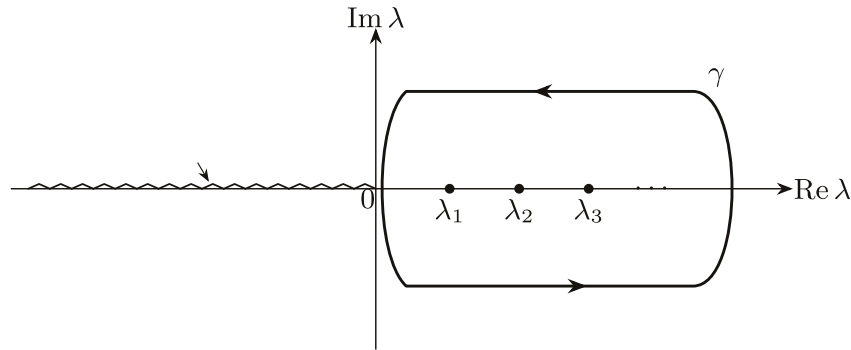


FIGURE 1. Schematic reconstruction of contour γ in the λ -plane.

Let A have eigenvalues λ_n and eigenfunctions ϕ_n :

$$A\phi_n = \lambda_n\phi_n. \quad (3.2)$$

Associate to it the zeta function

$$\begin{aligned} \zeta_A(s) &\equiv \text{Tr} \left[\frac{1}{A^s} \right] \\ &= \sum_{k=1}^{\infty} \lambda_k^{-s}. \end{aligned} \quad (3.3)$$

1. Show that zeta regularisation gives

$$\det A = \exp[-\zeta'_A(0)]. \quad (3.4)$$

Because ζ_A and its derivative need not converge near $s = 0$, analytic continuation may be required before this expression is meaningful.

Solution.

$$\det A = \prod_{k=1}^{\infty} \lambda_k, \quad (3.5)$$

while

$$\zeta'_A = \sum_{k=1}^{\infty} -\log \lambda_k \lambda^{-s}. \quad (3.6)$$

Therefore,

$$\det A = \exp[-\zeta'_A(0)]. \quad (3.7)$$

2. Take

$$A = -\frac{d^2}{d\tau^2} \quad (3.8)$$

on $0 \leq \tau \leq 1$, with Dirichlet conditions

$$\phi_n(0) = \phi_n(1) = 0. \quad (3.9)$$

Find the eigenvalues λ_n , then evaluate $\det A$ using

$$\zeta(0) = -\frac{1}{2}, \quad \zeta'(0) = -\frac{1}{2} \log(2\pi). \quad (3.10)$$

Solution. It can be easily found that $\lambda_n = n^2 \pi^2$, $n \in \mathbb{Z}^+$. Then,

$$\zeta_A(s) = \sum_{k=1}^{\infty} (n\pi)^{-2s}, \quad (3.11)$$

and

$$\zeta'_A(0) = \sum_{k=1}^{\infty} -2 \log k - 2 \log \pi = 2\zeta'(0) - 2 \log \pi \zeta(0) = -\log 2. \quad (3.12)$$

Hence, we have

$$\det A = 2. \quad (3.13)$$

In cases where the eigenvalues cannot be obtained explicitly, suppose one can construct $F(\lambda)$ whose zeros are precisely the eigenvalues of A .

3. Assume that the $\lambda_n > 0$ are discrete and nondegenerate. Show that

$$\zeta_A(s) = \frac{1}{2\pi i} \int_{\gamma} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda}, \quad (3.14)$$

where γ is the contour specified in Figure 1 and the branch cut lies on the negative real axis.

Solution. The RHS is given by

$$\begin{aligned}
& \frac{1}{2\pi i} \int_{\gamma} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda} \\
&= \sum_{n=1}^{\infty} \lim_{\lambda \rightarrow \lambda_n} \frac{\lambda - \lambda_n}{\lambda^s} \frac{1}{F(\lambda)} \frac{dF(\lambda)}{d\lambda} \\
&= \sum_{n=1}^{\infty} \lim_{\lambda \rightarrow \lambda_n} \left(F'(\lambda_n) + F''(\lambda_n)(\lambda - \lambda_n) + \dots \right) \left(F'(\lambda_n)(\lambda - \lambda_n) + \frac{1}{2} F''(\lambda_n)(\lambda - \lambda_n)^2 + \dots \right)^{-1} \\
&= \sum_{n=1}^{\infty} \lim_{\lambda \rightarrow \lambda_n} \frac{1}{\lambda^s} \left(1 + \frac{F''(\lambda_n)}{F'(\lambda_n)}(\lambda - \lambda_n) + \dots \right) \left(1 - \frac{1}{2} \frac{F''(\lambda_n)}{F'(\lambda_n)}(\lambda - \lambda_n) + \dots \right) \\
&= \sum_{n=1}^{\infty} \lambda_n^{-s} \\
&= \zeta_A(s).
\end{aligned} \tag{3.15}$$

4. Deform γ into the contour γ_- , which hugs the negative real axis as indicated in Figure 2, and derive

$$\zeta_A(s) = \frac{\sin(\pi s)}{\pi} \int_0^{-\infty} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda}. \tag{3.16}$$

Solution.

$$\begin{aligned}
\zeta_A(s) &= \frac{1}{2\pi i} \int_{\gamma_-} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda} \\
&= \lim_{\epsilon \rightarrow 0^+} \frac{1}{2\pi i} \int_{-\infty}^0 d\lambda (\lambda + i\epsilon)^{-s} \frac{d \ln F(\lambda + i\epsilon)}{d\lambda} - \frac{1}{2\pi i} \int_{-\infty}^0 d\lambda (\lambda - i\epsilon)^{-s} \frac{d \ln F(\lambda - i\epsilon)}{d\lambda} \\
&= \frac{1}{2\pi i} \int_{-\infty}^0 *0 d\lambda (e^{-s \log \lambda - i s \pi} - e^{-s \log \lambda + i s \pi}) \frac{d \ln F(\lambda)}{d\lambda} \\
&= -\frac{\sin \pi s}{\pi} \int_{-\infty}^0 d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda}.
\end{aligned} \tag{3.17}$$

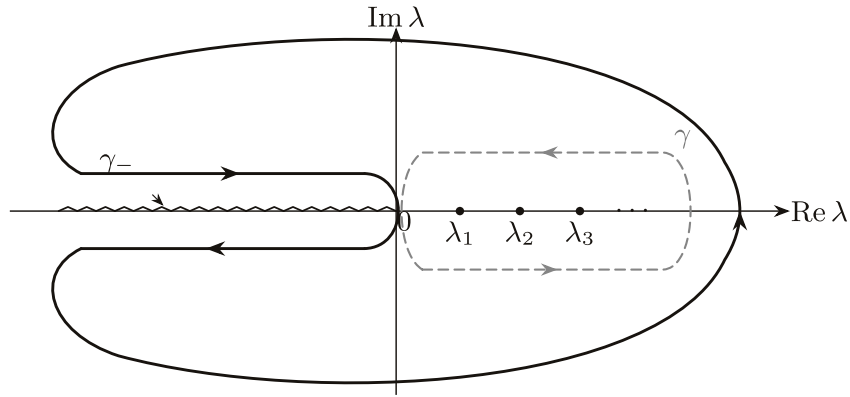


FIGURE 2. Finite contour deformation.

5. Apply the preceding results to the one-dimensional Schrödinger operator

$$A = -\frac{d^2}{dx^2} + V(x) \tag{3.18}$$

on $0 \leq x \leq 1$, with Dirichlet boundary conditions:

$$\begin{aligned} A\phi_n &= \lambda_n \phi_n, \\ \phi(0) &= 0, \\ \phi(1) &= 0. \end{aligned} \tag{3.19}$$

Assume that the λ_n are positive, discrete, and nondegenerate. Instead of the Dirichlet eigenfunctions, solve the same eigenvalue equation with initial-value conditions

$$\begin{aligned} Au_\lambda &= \lambda u_\lambda, \\ u_\lambda(0) &= 0, \\ u'_\lambda(0) &= 1. \end{aligned} \tag{3.20}$$

Choose

$$F(\lambda) = u_\lambda(1) \tag{3.21}$$

and use the previous parts to establish the Gel'fand--Yaglom theorem

$$\frac{\det A}{\det A_{\text{free}}} = \frac{u_0(1)}{u_{\text{free},0}(1)}, \tag{3.22}$$

where

$$A_{\text{free}} = -\frac{d^2}{dx^2}. \tag{3.23}$$

Solution. From the results above, we have

$$\zeta'_A(0) = \int_0^{-\infty} d\lambda \frac{d \log F(\lambda)}{d\lambda} = \log \frac{F(-\infty)}{F(0)} \implies \det A = \exp[-\zeta'_A(0)] = \frac{F(0)}{F(-\infty)}. \tag{3.24}$$

Note that $F(-\infty) = u_{-\infty}(1)$ is formally infinite. Intuitively, one might guess

$$\lim_{\lambda \rightarrow -\infty} \frac{F(\lambda)}{F_{\text{free}}(\lambda)} = 1, \tag{3.25}$$

which follows

$$\frac{\det A}{\det A_{\text{free}}} = \frac{u_0(1)}{u_{\text{free},0}(1)}. \tag{3.26}$$

The guess is true, as long as $V(x)$ is bounded, when $\lambda \rightarrow -\infty$, $V(x)$ becomes negligible.

6. Use the Gel'fand--Yaglom theorem to compute $\det A$ for

$$A = -\frac{d^2}{dx^2} + \omega^2 \tag{3.27}$$

on the unit interval with Dirichlet boundary conditions. Verify the result independently by finding the eigenvalues and using

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right). \tag{3.28}$$

Solution.

$$u_{\text{free},0}(x) = x, \quad u_0(x) = \frac{\sinh \omega x}{\omega}, \quad \det A_{\text{free}} = 2 \implies \det A = 2 \frac{\sinh \omega}{\omega}. \quad (3.29)$$

Meanwhile,

$$\lambda_n = \pi^2 n^2 + \omega^2 \implies \det A = \prod_{n=1}^{\infty} \pi^2 n^2 \left(1 + \frac{\omega^2}{\pi^2 n^2} \right), \quad (3.30)$$

which looks disastrous. However, remember, we can regulate it with

$$\det A_{\text{free}} = \prod_{n=1}^{\infty} \pi^2 n^2. \quad (3.31)$$

Therefore, we find

$$\frac{\det A}{\det A_{\text{free}}} = \frac{\sinh \omega}{\omega}, \quad (3.32)$$

which recovers the Fel'danf-Yaglom theorem result by regulating $\det A_{\text{free}}$ through the analytic continuation of the Reimann function.